

Live test on a real phone

streaming phone sensors with phyphox and running our maths in real time (Synheart Kinematics)

Short note on the live demo: instead of only testing on saved datasets, I streamed the motion sensor off my own phone over Wi-Fi and ran the Activity-State / Movement-Regularity maths on it *live*.

1 Setup

- Phone and laptop on the same Wi-Fi. The phone runs **phyphox** (*Acceleration with g*) with **Remote Access** on — this makes the phone serve its sensor data over a small web address, e.g. `http://<ip>:8080`.
- The laptop asks the phone for new samples every 0.1 s (the phyphox `/get` endpoint), keeps a **2-second sliding window**, and runs the rule.
- No trained model and no internet — our Activity State is a deterministic threshold rule, so the laptop just computes it directly.

2 What it computes (live, every ~ 0.4 s)

From the three sensor axes x, y, z :

$$|a| = \sqrt{x^2 + y^2 + z^2} \quad (\text{strength of acceleration}), \quad \text{movement} = \text{std}(|a|) \text{ over the window.}$$

- $|a|$ — at rest this is just gravity (≈ 9.8); motion adds to it. Rotating the phone changes the axes but *not* $|a|$ (same length), so it is orientation-proof.
- **movement** — how much $|a|$ wobbles over 2 s = how much the person is moving. This single number drives the verdict.
- **Activity State** — a plain rule: $\text{movement} < 1.2 \Rightarrow \text{still}$, $1.2-4 \Rightarrow \text{walking}$, $> 4 \Rightarrow \text{running/vigorous}$.
- **Movement Regularity** — whether the wobble repeats (autocorrelation): *calm* (still), *rhythmic* (steady walk/run), *erratic* (jerky).

3 What I observed

The verdict tracked what I actually did, with clean transitions (Fig. 1):

what I did	movement (m/s ²)	$ a $	verdict
held still	0.01–0.02	≈ 9.86	STILL (calm)
walked	1.5–3.6	10–11	WALKING (rhythmic)
ran / shook	9–14	14–20	RUNNING/VIGOROUS (rhythmic)
stopped	$\rightarrow 0.01$	≈ 9.86	STILL (calm)

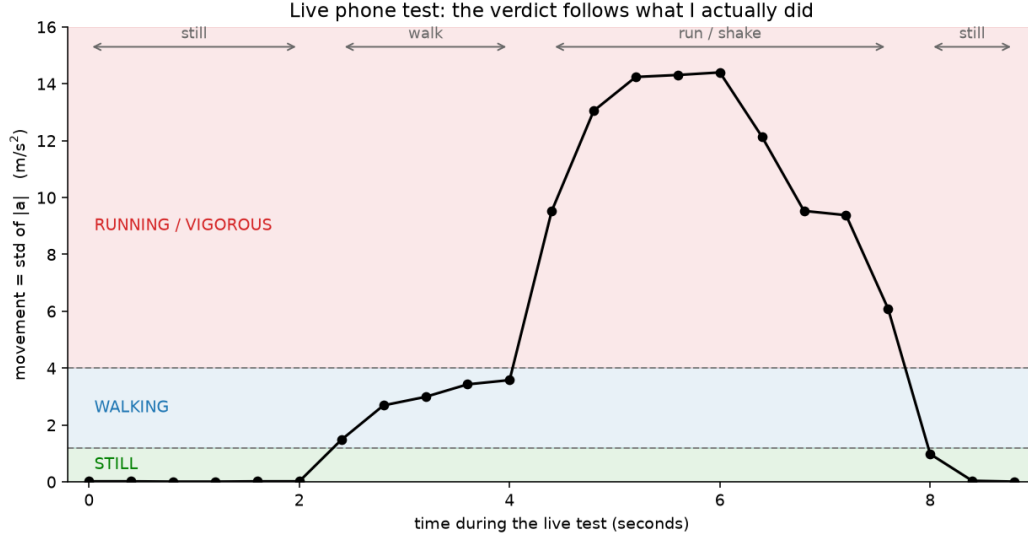


Figure 1: Real values captured during the live test. **movement** (one number from $|a|$) rises and falls exactly with what I did; the dashed lines are the still/walking/running thresholds.

The orientation-proof moment. Holding the phone still and rotating it to a different orientation: the per-axis numbers swing (gravity moves between x, y, z), but $|a|$, **movement** and the verdict stay put. Orientation changed completely; the activity reading did not — because $\|Ra\| = \|a\|$.

4 What this shows

- Both deliverables — **Activity State** and **Movement Regularity** — run **live, in real time, on a brand-new device** (a phone, not the training data), straight from the raw sensor.
- It stays **orientation-proof** on real hardware, not just on paper.
- No sitting/standing/lying: that is posture, which needs gravity *direction* (orientation-dependent) and belongs to the separate Postural-State task — by design.

This is the real-world face-check on top of the measured results: **0% drop under rotation on UCI-HAR** and **89% on unseen people in PAMAP2**. Measured on two datasets, and now running live on my own phone.